Assessments

MATH 107: Introduction to Linear Algebra — Fall 2026

Labs

Lab	Date	Learning objectives
Lab 0: Computers Speak Vectors	Fri Aug 28	Form and interpret linear combinations; enter, index, and modify vectors in Octave or Python; plot and debug supplied code.
Lab 1: Finding Similar Things	Fri Sep 18	Load vector data; compute distances; identify neighbors or clusters; hand-check a subset; interpret in context.
Lab 2: Transforming Space	Fri Oct 9	Predict and apply 2D/3D matrix transformations; plot before/after; adapt a short program.
Lab 3: Predict a System Through Time	Fri Oct 23	Implement $x_{t+1} = Ax_t$; iterate; hand-check first steps; compare parameters; record unexplained long-run behavior.
Micro-lab: computational cost	Fri Oct 30	Compare equivalent computation orders; count storage; measure runtime; justify an order quantitatively.
Lab 4: Why Did the System Behave That Way?	Fri Nov 20	Compute eigenpairs with software; verify $Av \approx \lambda v$; explain a simulation; interpret complex eigenvalues.
Lab 5: Impossible Equations	Wed Dec 2	Compute a least-squares fit; plot residuals and RMS; check orthogonality to columns of A; interpret usefulness.
Lab 6: Capstone	Fri Dec 4 – Wed Dec 9	Low-rank strategy, error/storage/runtime, and communication

Quizzes

Ten quizzes, in the first 10 minutes of class. The lowest two scores are dropped. Practice packets are grouped as **Core** and **Stretch**.

A consistent quiz has three parts:

- **Fluency** (4–5 points): one small calculation closely modeled on a practice item.
- **Meaning** (3–4 points): explain or interpret what the calculation says.
- **Computation/application** (2–3 points): interpret code, a plot, a residual, or software output.

You will not write substantial code from scratch on a timed quiz. Typical prompts: complete one missing line, predict three lines of output, identify an error, interpret output, or choose an appropriate command.

Every second or third quiz includes a short cumulative item from an earlier packet (beginning with Quiz 3).

Source key

Code	Source
B 3.10	Boyd and Vandenberghe, Exercise 3.10
S 6.1 #10	Strang, Section 6.1, Problem 10
L Ch. 2 Ex. 3	Lachniet, *Introduction to GNU Octave* (3rd ed., 2020), Chapter 2, Exercise 3
C	Course-created bridge or lab checkpoint

A "prompt" can be one selected part of a multi-part textbook exercise.

CLO coverage

Official course learning outcomes are assessed more than once; nothing important is a single-day checklist item.

Required outcome	Where it is developed and assessed
Interpret vectors, matrices, and complex numbers algebraically and geometrically	Weeks 1–3, 7, and 13
Solve systems by row reduction by hand and with software	Week 8; Midterm 2; cumulative final
Find 2×2 eigenpairs by hand and general eigenpairs with software	November 18 and Lab 4
Apply eigenvalues and eigenvectors	Dynamics in Week 9, explained in Week 13; Markov/normal-mode applications
Recognize triangular, symmetric, and positive-definite matrices	October 5, November 6, and December 2
Use linear combination, subspace, span, independence, basis, and dimension	Weeks 1 and 5, revisited in Weeks 8 and 11

Required outcome	Where it is developed and assessed
Write or adapt programs involving matrices, vectors, and plotting	Labs 0–6, with steadily reduced scaffolding
Apply linear algebra to real-world problems	Every lab and most class examples
Learn independently and collaboratively	Practice banks, low-stakes quizzes, paired work, lab studio, and capstone presentation
ILO/GEAR quantitative reasoning in context	Interpretation questions throughout; least-squares and capstone conclusions

Quiz dates

Quiz	Date	Focus
Quiz 1	Fri Sep 4	Vectors, linear combinations, Lab 0
Quiz 2	Fri Sep 11	Dot products and linear functions
Quiz 3	Fri Sep 18	Norm, distance, angle, and similarity
Quiz 4	Mon Sep 28	Clustering, dependence, span, basis, and dimension
Quiz 5	Mon Oct 12	Matrices, Ax, and transformations
Quiz 6	Mon Oct 19	Systems, elimination, and RREF
Quiz 7	Mon Oct 26	Dynamics and Lab 3
Quiz 8	Fri Nov 6	Matrix multiplication, cost, and inverses
Quiz 9	Mon Nov 30	Eigenvalues, complex numbers, Lab 4
Quiz 10	Fri Dec 4	Least squares, residuals, and Lab 5

There is no quiz on rank/determinants the week of Midterm 2; use the Midterm 2 practice packet. SVD is not on Quiz 10; use the capstone practice packet.

Exams

In-class, for the entire class period.

Exam	Date	Learning objectives
Midterm 1	Fri Oct 2	Compute with vectors and linear functions; interpret operations; analyze distance, similarity, span, and independence; select and communicate a method.
Midterm 2	Fri Nov 13	Analyze matrices as transformations and models; solve and interpret systems; use products, powers, rank, inverses, and determinants; interpret computational results.
Final exam	Wed Dec 16, 10:20 a.m.–12:10 p.m., FR 201	Hand fluency; select tools in new contexts; interpret software; explain connections among major concepts; apply linear algebra to quantitative problems.

Project

Project	Dates	Learning objectives
Capstone	Dec 4 launch, Dec 7 studio, Dec 9 gallery	Low-rank SVD strategy; compare reconstruction error, storage, and runtime; present quantitative evidence and limitations.

Class participation

Attendance and participation in class.